

Equivalent Martingale Measures

From the Probability Space to the T-Forward Measure

A self-contained derivation with two-state and continuous-time examples

1. The Probability Space

We work on a **filtered probability space** $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions:

- Ω — the *sample space*: every possible path the world can take. In FX, each $\omega \in \Omega$ is one trajectory of the spot rate $S(t)$ from today to maturity.
- $\mathcal{F}$ — the *sigma-algebra* of all observable events.
- $\{\mathcal{F}_t\}$ — the *filtration*: $\mathcal{F}_t$ is the information available up to time t . At $t = 0$ you know only $S(0)$; at time t you know $\{S(u) : u \leq t\}$. The filtration grows: $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s < t$.
- $\mathbb{P}$ — the *real-world* (physical) probability measure. It describes what actually happens in markets, including risk premia.

Key intuition. Ω is fixed — it is the collection of all paths. A *measure* is simply an assignment of probabilities to events (sets of paths). Different measures assign different weights to the same paths. Changing measure does *not* change which paths exist, only how likely each one is considered to be.

2. Numeraires and the EMM

Definition 2.1 (Numeraire). A **numeraire** is any strictly positive $\mathcal{F}_t$ -adapted price process $N(t) > 0$. It serves as the unit of account: all asset prices are expressed relative to it.

Common choices of numeraire:

Numeraire $N(t)$	Description	Associated measure
$B(t) = e^{r_d t}$	Money-market account	Risk-neutral $\mathbb{Q}$
$P(t, T)$	T-maturity zero-coupon bond	T-forward measure $\mathbb{Q}^T$
$S(t) e^{r_f t}$	Foreign bank account	Foreign risk-neutral $\mathbb{Q}^f$

Definition 2.2 (Equivalent Martingale Measure (EMM)). A probability measure $\mathbb{Q} \sim \mathbb{P}$ is an **EMM** associated with numeraire N if every asset price $V(t)$, after dividing by N , is a local $\mathbb{Q}$ -martingale:

$$\frac{V(t)}{N(t)} = \mathbb{E}^{\mathbb{Q}} \left[V(T) \mid \mathcal{F}_{N(T)/t} \right] \iff \frac{V(t)}{N(t)} = \mathbb{E}^{\mathbb{Q}} \left[\frac{V(T)}{N(T)} \mid \mathcal{F}_t \right] \quad (1)$$

2.1. Why divide by the numeraire?

Raw prices are not martingales — they drift (stocks earn risk premia, bonds accrue interest). But *relative prices* V/N are. The choice of N determines which drift vanishes.

With $N = B(t) = e^{r_d t}$:

$$\frac{V(t)}{B(t)} = \mathbb{E}^{\mathbb{Q}} \left[\frac{V(T)}{B(T)} \middle| \mathcal{F}_t \right] \implies V(t) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r_d(T-t)} V(T) \middle| \mathcal{F}_t \right]$$

This is the **risk-neutral pricing formula**: price = discounted expected payoff under $\mathbb{Q}$.

With $N = P(t, T)$:

$$\frac{V(t)}{P(t, T)} = \mathbb{E}^{\mathbb{Q}^T} \left[\frac{V(T)}{P(T, T)} \middle| \mathcal{F}_t \right] = \mathbb{E}^{\mathbb{Q}^T} [V(T) | \mathcal{F}_t]$$

since $P(T, T) = 1$. No discounting inside the expectation — this is the **T-forward pricing formula**.

3. Deriving $\mathbb{Q}$ in a Two-State World

We make things concrete with the simplest possible model.

3.1. Setup

- Today $t = 0$, maturity $T = 1$, risk-free rate $r = 5\%$, so $B(1) = 1.05$.
- Stock price $S(0) = 100$, with two possible outcomes:

$$S(1) = \begin{cases} 120 & \text{state } \omega_u \text{ (up), with } \mathbb{P}\text{-probability } p = 0.70 \\ 80 & \text{state } \omega_d \text{ (down), with } \mathbb{P}\text{-probability } 1 - p = 0.30 \end{cases}$$

3.2. Finding $\mathbb{Q}$

Let $\mathbb{Q} = (q, 1 - q)$ be the risk-neutral probabilities. We impose the EMM condition with numeraire B :

$$\frac{S(0)}{B(0)} = \mathbb{E}^{\mathbb{Q}} \left[\frac{S(1)}{B(1)} \middle| \mathcal{F}_0 \right]$$

Since $B(0) = 1$:

$$100 = q \cdot \frac{120}{1.05} + (1 - q) \cdot \frac{80}{1.05}$$

Multiply both sides by 1.05:

$$105 = 120q + 80(1 - q) = 80 + 40q$$

$$q = \frac{105 - 80}{40} = \frac{25}{40} = 0.625 \implies \mathbb{Q} : q_u = 0.625, \quad q_d = 0.375$$

Remark 3.1. The real-world probability $p = 0.70$ played no role. $\mathbb{Q}$ is determined entirely by $S(0)$, r , S_u , and S_d . In general:

$$q = \frac{B(1) \cdot S(0) - S_d}{S_u - S_d} = \frac{1.05 \times 100 - 80}{120 - 80} = 0.625$$

This is the **risk-neutral probability**: the unique q that makes the discounted stock price a martingale.

3.3. Pricing a Derivative Under $\mathbb{Q}$

Price a call option with strike $K = 100$:

$$\text{Payoff}(\omega_u) = \max(120 - 100, 0) = 20, \quad \text{Payoff}(\omega_d) = 0$$

$$V(0) = \frac{1}{B(1)} \mathbb{E}^{\mathbb{Q}}[\text{Payoff} | \mathcal{F}_0] = \frac{1}{1.05} [0.625 \times 20 + 0.375 \times 0] = \frac{12.50}{1.05} \approx 11.90$$

4. Change of Measure and the Radon-Nikodym Derivative

4.1. The Radon-Nikodym Derivative

The bridge between two equivalent measures $\mathbb{Q}$ and $\mathbb{Q}^T$ is the **Radon-Nikodym derivative** $\frac{d\mathbb{Q}^T}{d\mathbb{Q}}$. It is a random variable (one value per state ω) that reweights probabilities:

$$\mathbb{Q}^T(A) = \mathbb{E}^{\mathbb{Q}} \left[\frac{d\mathbb{Q}^T}{d\mathbb{Q}} \cdot \mathbf{1}_A \right] \quad \text{for all events } A \in \mathcal{F}$$

The general formula when changing from numeraire N (measure $\mathbb{Q}$) to numeraire N^T (measure $\mathbb{Q}^T$) is:

$$\left. \frac{d\mathbb{Q}^T}{d\mathbb{Q}} \right|_{\mathcal{F}_T} = \frac{N^T(T)/N^T(0)}{N(T)/N(0)}$$

4.2. Two-State Example: $\mathbb{Q} \rightarrow \mathbb{Q}^T$ (T-Bond Numeraire)

Use $N = B$ (money-market, measure $\mathbb{Q}$) and $N^T = P(t, T)$ (T-bond, measure $\mathbb{Q}^T$).

With flat deterministic rates: $P(0, 1) = 1/1.05 = 0.952$ and $P(1, 1) = 1$ in both states.

$$\frac{d\mathbb{Q}^T}{d\mathbb{Q}}(\omega_u) = \frac{P(1, 1)/P(0, 1)}{B(1)/B(0)} = \frac{1/0.952}{1.05/1} = \frac{1.05}{1.05} = 1$$

$$\frac{d\mathbb{Q}^T}{d\mathbb{Q}}(\omega_d) = 1$$

So $\mathbb{Q}^T = \mathbb{Q}$ with deterministic rates. The T-forward measure coincides with the risk-neutral measure when rates are non-stochastic.

4.3. Two-State Example: $\mathbb{Q} \rightarrow \mathbb{Q}^S$ (Stock Numeraire)

Now use $N^S = S(t)$ as numeraire (the “stock measure” $\mathbb{Q}^S$). The Radon-Nikodym derivative from $\mathbb{Q}$ to $\mathbb{Q}^S$:

$$\frac{d\mathbb{Q}^S}{d\mathbb{Q}}(\omega_u) = \frac{S(1)/S(0)}{B(1)/B(0)} = \frac{120/100}{1.05} = \frac{1.20}{1.05} = 1.143$$

$$\frac{d\mathbb{Q}^S}{d\mathbb{Q}}(\omega_d) = \frac{80/100}{1.05} = \frac{0.80}{1.05} = 0.762$$

Updating probabilities:

$$q_u^S = q_u \cdot 1.143 = 0.625 \times 1.143 = 0.714$$

$$q_d^S = q_d \cdot 0.762 = 0.375 \times 0.762 = 0.286$$

State	$\mathbb{P}$	$\mathbb{Q}$	$\mathbb{Q}^S$
ω_u (up)	0.700	0.625	0.714
ω_d (down)	0.300	0.375	0.286

Under $\mathbb{Q}^S$ the up state is more likely than under $\mathbb{Q}$. Intuitively: from the perspective of someone holding the stock, good outcomes (high stock prices) receive more weight. This is the essence of change of measure — it tilts probabilities toward the outcomes that matter for the new numeraire.

5. Continuous Time: Girsanov’s Theorem

In continuous time the Radon-Nikodym derivative becomes a **stochastic exponential**.

5.1. Setup Under $\mathbb{Q}$

Under the risk-neutral measure $\mathbb{Q}$ (numeraire $B(t) = e^{r_d t}$), the FX spot dynamics are:

$$\frac{dS}{S} = (r_d - r_f) dt + \sigma dW^{\mathbb{Q}}$$

The forward $F(t, T) = S(t) e^{(r_d - r_f)(T - t)}$ satisfies:

$$\frac{dF}{F} = \sigma dW^{\mathbb{Q}}$$

F is already driftless under $\mathbb{Q}$. We want to make this rigorous by changing to $\mathbb{Q}^T$.

5.2. Girsanov’s Theorem

Theorem 5.1 (Girsanov). Let $\theta(t)$ be an adapted process satisfying the Novikov condition $\mathbb{E}[e^{\frac{1}{2} \int_0^T \theta^2(t) dt}] < \infty$. Define the Radon-Nikodym derivative:

$$\left. \frac{d\mathbb{Q}^T}{d\mathbb{Q}} \right|_{\mathcal{F}_T} = \exp\left(\int_0^T \theta(t) dW^{\mathbb{Q}} - \frac{1}{2} \int_0^T \theta^2(t) dt\right) =: \mathcal{E}(\theta \cdot W)_T$$

Then the process $W^{\mathbb{Q}^T}(t) = W^{\mathbb{Q}}(t) - \int_0^t \theta(s) \, ds$ is a standard Brownian motion under $\mathbb{Q}^T$.

The drift adjustment θ is the **market price of risk** for the change of numeraire.

5.3. Computing the T-Forward Measure

To change from $\mathbb{Q}$ (numeraire B) to $\mathbb{Q}^T$ (numeraire $P(t, T)$), we need the volatility σ_P of the T-bond. With **deterministic flat rates**:

$$P(t, T) = e^{-r_d(T-t)} \implies \sigma_P = 0 \implies \theta = 0$$

So:

$$dW^{\mathbb{Q}^T} = dW^{\mathbb{Q}} \quad (\text{flat rates})$$

The forward dynamics are identical under $\mathbb{Q}$ and $\mathbb{Q}^T$:

$$\frac{dF}{F} = \sigma \, dW^{\mathbb{Q}^T}$$

5.4. General Case: Stochastic Rates

With stochastic rates (e.g. Hull-White model), $P(t, T)$ is itself stochastic with volatility $\sigma_P(t)$. Then:

$$\theta(t) = \sigma_P(t)$$

and the Brownian motion shifts by:

$$dW^{\mathbb{Q}^T} = dW^{\mathbb{Q}} - \sigma_P(t) \, dt$$

The forward dynamics under $\mathbb{Q}^T$ become:

$$\frac{dF}{F} = \sigma_F \, dW^{\mathbb{Q}^T} \quad (\text{driftless, as required})$$

5.5. The Radon-Nikodym Derivative Explicitly

For constant θ , the Radon-Nikodym derivative is:

$$\left. \frac{d\mathbb{Q}^T}{d\mathbb{Q}} \right|_{\mathcal{F}_T} = \exp\left(\theta W^{\mathbb{Q}}(T) - \frac{1}{2}\theta^2 T\right)$$

The expectation under the new measure is computed as:

$$\mathbb{E}^{\mathbb{Q}^T}[X \mid \mathcal{F}_t] = \frac{1}{\mathcal{E}(\theta \cdot W)_t} \mathbb{E}^{\mathbb{Q}}[X \cdot \mathcal{E}(\theta \cdot W)_T \mid \mathcal{F}_t]$$

6. Summary

Measure	Numeraire	BM shift θ	What is driftless
Real world $\mathbb{P}$	—	—	Nothing useful for pricing
Risk-neutral $\mathbb{Q}$	$B(t) = e^{r_d t}$	Removes risk premium	$V(t)/B(t)$
T-forward $\mathbb{Q}^T$	$P(t, T)$	$+\sigma_P(t)$	$F(t, T)$
Stock $\mathbb{Q}^S$	$S(t)$	$+\sigma$	$V(t)/S(t)$
Foreign $\mathbb{Q}^f$	$S(t)e^{r_f t}$	$+\sigma$	Foreign-discounted prices

The chain of reasoning is always:

1. Choose a numeraire N convenient for the problem.
2. Find the associated EMM $\mathbb{Q}^N$ via the Radon-Nikodym derivative.
3. Identify the new Brownian motion $W^{\mathbb{Q}^N}$ via Girsanov (add drift $\theta = \sigma_N$ to old BM).
4. Price as $V(0) = N(0) \mathbb{E}^{\mathbb{Q}^N}[V(T)/N(T)]$ — no other discounting needed.